

VERIFIED EXECUTABLE LINEAR ALGEBRA

DenseMatrix makes Lean matrix algorithms run like data.

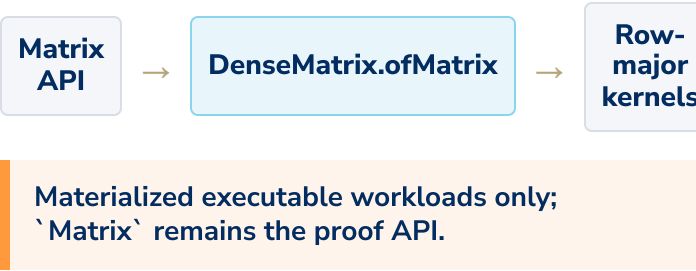
Row-major data for RREF, LU, and determinant kernels, bridged back to proof-friendly `Matrix` theorems.

TEAM
Annis Wu · Joseph Qian · Junye Ji
· Veer Shukla
Mentor: Dhruv Bhatia



Why DenseMatrix

DenseMatrix stores entries in row-major arrays, so matrix code can execute over contiguous data instead of repeatedly re-evaluating pointwise functions.



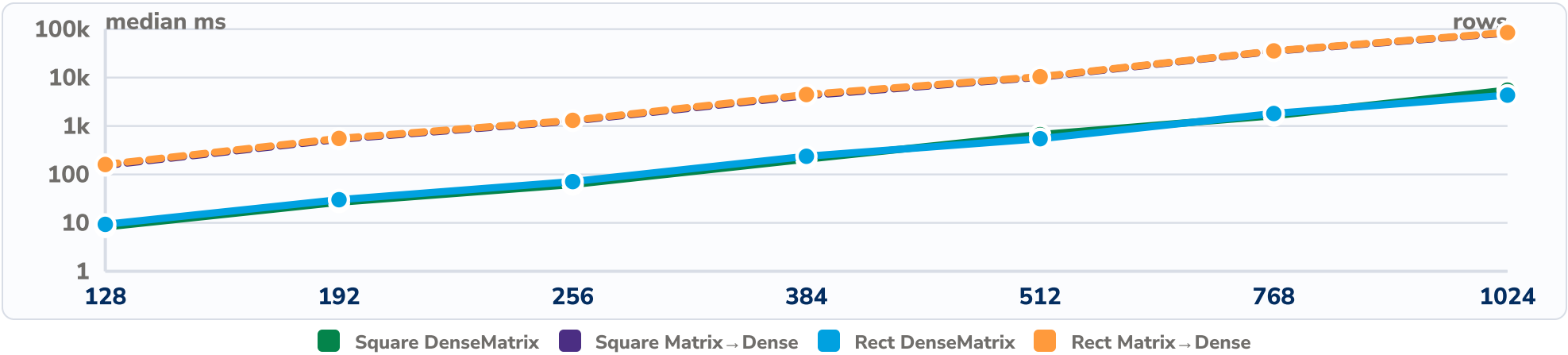
Proof Connection

- DenseMatrix.toMatrix_add
- DenseMatrix.toMatrix_mul
- rowEchelonForm_toMatrix
- luDet_eq_det
- gaussDet_eq_det

Executable backend results stay connected to the mathematical API by theorem anchors.

Multiplication Scaling

DenseMatrix vs Matrix → Dense, median ms on a log scale.



Representation Tradeoff

Pointwise primitives at the largest complete size.

OPERATION	DENSE	MATRIX	MATRIX → DENSE	WINNER
Add	54.0	45.1	47.5	Matrix
Scalar	53.2	30.3	35.3	Matrix
Transpose	55.3	43.5	46.8	Matrix

Benchmark Scope

2247 RECORDS	328 MATERIALIZED ROWS	2048 MAX ROWS
Protocol	30 repeats · 5 warmups	
Sizes	16, 32, 64, 96, 128, 192, 256	
Stress	384, 512, 768, 1024	
Machine	pinned core · environment archived	

Algorithm Kernels

Representative Rat kernels at n=128.

REF	n=128	16292 ms
RREF	n=128	32394 ms
LU factor	n=128	17090 ms
Gauss det	n=128	16234 ms
LU det	n=128	16933 ms